

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	CSA0609	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 3 – Comparative Analysis
Weightage:	10% of CIA	Total Marks:	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
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Section / Batch:	IT	Date:	25/08/2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze the given questions thoroughly before answering.
- Compare multiple aspects such as methods, results, advantages, and limitations clearly.
- Critically evaluate the strengths, weaknesses, and implications with proper justification.
- Present meaningful insights, interpretations, and well-supported conclusions from the comparisons.

Question Type	Focus Area	Questions
Comparative Analysis	Comparative analysis of Dynamic Programming and Greedy algorithms for optimization problems.	Q1

SECTION A – Comparative Analysis

Code of Conduct

I S Bhuvanesh (192521004) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S .Bhuvanesh

RUBRICS FOR CALCULATION

Comparative Analysis					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%				
Comparison & Differentiation	25%				
Critical Evaluation	25%				
Synthesis & Insight Generation	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

ASSESSMENT TOOL 3 – Comparative Analysis

Q5. Activity Selection (Greedy vs DP)

Given activities with start times [1, 3, 0, 5, 8, 5] and finish times [2, 4, 6, 7, 9, 9], solve using both Greedy and Dynamic Programming approaches. Compare selected activities and efficiency. Evaluate correctness of greedy choice property. Derive insights on scheduling optimization. Instructions: Solve the assigned problem, show the algorithm/steps clearly, analyze correctness, time and space complexity, and provide the requested comparison or insights.

1. INTRODUCTION

The Activity Selection Problem is a classic scheduling problem in which we have a set of activities, and each activity has a start time and a finish time. The objective is to select the maximum number of activities that do not overlap.

For this problem, we solve the same input using two approaches:

- Greedy Algorithm
- Dynamic Programming (DP)

Then, we compare their selected activities, correctness, time complexity, space complexity, and efficiency.

2. PROBLEM STATEMENT

Given activities with:

- Start times: [1, 3, 0, 5, 8, 5]
- Finish times: [2, 4, 6, 7, 9, 9]

The objective is to select the maximum number of non-overlapping activities using:

1. Greedy approach
2. Dynamic Programming approach

2. ACTIVITY TABLE

Activity	Start	Finish
A1	1	2
A2	3	4
A3	0	6
A4	5	7
A5	8	9
A6	5	9

3. GREEDY APPROACH

3.1 Algorithm

1. Sort activities according to increasing finish time.
2. Select the first activity.
3. For every next activity, select it if its start time is greater than or equal to the finish time of the previously selected activity.
4. Continue until all activities are checked.

3.2 Steps

Activities are already sorted by finish time:

A1(1,2), A2(3,4), A3(0,6), A4(5,7), A5(8,9), A6(5,9)

- Select A1 (1,2)

- A2 starts at $3 \geq 2 \rightarrow$ Select A2
- A3 starts at $0 < 4 \rightarrow$ Reject
- A4 starts at $5 \geq 4 \rightarrow$ Select A4
- A5 starts at $8 \geq 7 \rightarrow$ Select A5
- A6 starts at $5 < 9 \rightarrow$ Reject

Greedy Result

Selected activities: A1, A2, A4, A5

Maximum number of activities = 4

4. DYNAMIC PROGRAMMING APPROACH

Idea

Dynamic Programming considers the best solution for smaller sets of activities and builds the final optimal solution from them.

For each activity, we determine the previous compatible activity and calculate:

$$DP[i] = \max(DP[i-1], 1 + DP[j])$$

where j is the last activity that finishes before activity i starts.

DP Calculation

After sorting by finish time:

Activity			Start	Finish	Best Count
A1	1	2	1		
A2	3	4	2		
A3	0	6	2		
A4	5	7	3		
A5	8	9	4		
A6	5	9	3		

Therefore:

DP optimum = 4 activities

One optimal selection is:

$A1 \rightarrow A2 \rightarrow A4 \rightarrow A5$

5. CORRECTNESS OF GREEDY CHOICE

The greedy choice is correct because selecting the activity with the earliest finishing time leaves the maximum possible remaining time for other activities.

For example:

- A1 finishes at 2.
- A2 finishes at 4.
- A4 finishes at 7.
- A5 finishes at 9.

This allows four activities to be scheduled.

Greedy Choice Property

The activity with the earliest finish time can always be chosen as part of an optimal solution. Replacing the first activity of an optimal solution with the earliest-finishing activity does not reduce the number of activities that can subsequently be selected.

Therefore, the greedy algorithm gives the optimal solution for the activity-selection problem.

6. COMPLEXITY ANALYSIS

Approach	Time Complexity	Space Complexity
Greedy	$O(n \log n)$	$O(n)$
Dynamic Programming	$O(n^2)$	$O(n)$

Greedy: Sorting requires $O(n \log n)$ and selecting activities requires $O(n)$.

DP: For every activity, compatible previous activities may need to be checked, resulting in $O(n^2)$ time.

7. COMPARATIVE ANALYSIS

Feature	Greedy	Dynamic Programming
Strategy	Earliest finish time	Compare multiple subproblems
Selected activities	A1, A2, A4, A5	A1, A2, A4, A5
Maximum activities	4	4
Time	$O(n \log n)$	$O(n^2)$
Space	$O(n)$	$O(n)$
Implementation	Simple	More complex
Optimal for this problem	Yes	Yes
Efficiency	Higher	Lower

8. GREEDY VS DYNAMIC PROGRAMMING

Feature	Greedy	Dynamic Programming
Main idea	Choose earliest finish	Compare possible choices
Result	4 activities	4 activities
Selected activities	A1, A2, A4, A5	A1, A2, A4, A5
Space	$O(n)$	$O(n)$
Implementation	Simple	More complex

Optimal for this problem	Yes	Yes
Best use	Standard activity selection	Problems with more complex constraints

9. ADVANTAGES OF GREEDY APPROACH

- Simple to implement
- Fast execution
- Requires less computation than basic DP.
- Gives an optimal solution for the standard Activity Selection problem.
- Suitable for large scheduling datasets.

10. ADVANTAGES OF DYNAMIC PROGRAMMING

- Considers multiple possible choices.
- Useful when greedy choice property does not exist.
- Can handle more complicated optimization problems.
- Provides a systematic way to construct an optimal solution.
- Can be extended to variants with additional constraints.

11. REAL-WORLD APPLICATIONS

Activity Selection is useful in:

- **Classroom Scheduling**
Selecting maximum classes that can use a classroom without time conflicts.
- **Meeting Scheduling**
Selecting the maximum number of meetings that can be conducted in one meeting room.
- **Job Scheduling**
Selecting compatible jobs for a single machine or worker.

➤ **Conference Planning**

Selecting maximum conference sessions without overlapping.

➤ **CPU Scheduling**

Selecting non-overlapping tasks when a resource can execute one task at a time

12. INSIGHTS ON SCHEDULING OPTIMIZATION

- Greedy is more efficient for the standard activity-selection problem.
- Choosing the earliest finishing activity maximizes the remaining scheduling time.
- Dynamic Programming also finds the optimal answer but performs more computations.
- Both approaches select 4 non-overlapping activities for the given input.
- Greedy is preferred when the problem satisfies the greedy-choice property.
- DP becomes useful when a problem has overlapping subproblems and does not have a simple greedy-choice property.

13. CONCLUSION

For the given activity-selection problem, both Greedy and Dynamic Programming produce the optimal result of 4 activities. The selected activities are A1, A2, A4, and A5. However, the Greedy method is more efficient, requiring $O(n \log n)$ time compared with $O(n^2)$ for Dynamic Programming. Hence, Greedy is the better approach for standard activity selection because its greedy-choice property guarantees optimal scheduling with lower computational cost.